

Software Engineering (CS383) - Project Report
Citizen Care System

Semester: 472

Due Date: Jun 3

Team members:

Nada Alharbi	432205291
Leen Almansour	432205435
Norah Almrzooq	441203264
Dhiaa Alharbi	432204322
Gader Alquniha	441203382
Rahma Alsuhaymi	432205312

Supervisor: Dr.Rafaa Aljarbou

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Introduction

The Citizens Care System is a modern digital platform designed to simplify access to public and citizen services through one centralized system. The project aims to provide users with a convenient and organized environment where they can manage services, book appointments, receive notifications, and access important information quickly and efficiently. The system was developed to reduce traditional procedures and improve communication between citizens and service providers through an easy-to-use interface.

In recent years, technology has played a major role in transforming government and public services into digital platforms that save time and effort for users. As a result, there is an increasing demand for smart systems that can provide secure, fast, and reliable services online. Based on this idea, the Citizens Care System was created to support digital transformation and enhance the overall user experience by offering multiple services within a single platform.

The project also applies important software engineering concepts such as requirements analysis, system design, database management, and UML diagrams. During the development process, different software engineering methodologies and design principles were considered to ensure that the system is functional, efficient, and user-friendly. In addition, teamwork and project management practices were used to organize tasks and complete the project successfully.

Furthermore, the system focuses on usability and accessibility to ensure that users from different backgrounds can interact with the platform smoothly. Security and data organization were also considered as important aspects of the project to provide a reliable and structured system environment.

Moreover, the Citizens Care System was inspired by the concept and services of the Saudi application Tawakkalna, which encouraged us to design a similar centralized platform with our own features, and project requirements.

Finally, this project provided us with valuable experience in teamwork and project organization. During the project, we improved our communication, problem-solving, and analytical thinking skills while working together to achieve the project goals. The Citizens Care System also increased our understanding of how digital platforms can support citizens and improve access to services in a more organized and efficient way.

Project Management

2.1 Group Meetings Documentation

Meeting	Date	Duration	Attendance	Main Topics	Minutes	Minutes writer
1	14.3.2026	1 hour	All members	Project topic selection, team roles, tool setup	topic confirmed as Citizen Care System, work starts , tasks assigned with deadlines	Leen
2	15.4.2026	2 hours	All members	Reviewing the remaining tasks, Discussing UML .diagrams	Expanded external interface requirements, ensured details are clear, and supported linking requirements to use cases.	Gader
3	17.5.2026	2 hours	All members	We practiced the presentation and ensured that the time stayed within the allowed limit.	Organized meeting content, monitored time, and helped finalize presentation and report.	Rahma
4	19.5.2026	1 hour	All members	Presentation fine ,tuning,repot final review ,and project submission working	Presentation order set, time management confirmed, training done	Nada

2.2 Repository Organization

The team maintained a centralized and organized repository using a shared canva document. This shared document served as the main workspace for all project, members and contained all project artifacts, including the SRS, SDD, UML descriptions, ethics section, and modeling documentation. It ensured real-time collaboration, consistent updates, and easy access for all group members throughout the project

3. Software Engineering Ethics and Responsibility

3.1 Ethical Principles Addressed in the Project

In software engineering, ethics play a crucial role in ensuring that the products developed are responsible, trustworthy, and beneficial to society. For an application like Citizen Care System, ethical considerations are paramount as it manages highly sensitive personal and health data that directly impact the safety and well-being of the entire community.

To develop this application in an ethical manner, we adhere to the core principles outlined by Software Engineering Ethics:

1. PRIVACY AND DATA PROTECTION

Since Citizen Care System

serves as a comprehensive national digital identity, the principle of Privacy is the fundamental pillar.

Commitment: Software engineers must ensure the encryption of all personal and health data (such as immunization status and travel history).

Application: Geographic locations and vital data are accessed only when strictly necessary (e.g., for permits) while ensuring that this information is not shared with unauthorized parties, in alignment with national data protection policies.

2. PUBLIC INTEREST AND SAFETY

The application is designed to serve the public interest, especially during health crises or for national security and convenience.

Commitment: The app must work efficiently and effectively during critical times, prioritizing the well-being of the users.

Application: By providing accurate health status information and warning against crowded areas, the app contributes to saving lives and reducing risks, ensuring that the public's safety is the priority in every programming decision.

3. PRODUCT QUALITY AND SECURITY

This principle emphasizes the importance of the quality and reliability of the digital product.

Commitment: The application must be reliable, efficient, and accurate, ensuring it functions as expected in urgent or official situations.

Application: High security standards, such as end-to-end encryption (TLS 1.3), are integrated to protect communications. This ensures that features like the "Digital Identity" work with 100% accuracy to prevent any legal or technical issues for users.

4. JUDGMENT AND INTEGRITY

Engineers involved in the development must exercise sound judgment when making decisions about design and functionality.

Commitment: Choices made should prioritize user safety and ensure the app performs its tasks without error.

Application: Continuous code reviews are conducted to ensure the system operates with integrity and transparency, especially in algorithms that manage service priorities or permit issuance, ensuring fairness for all segments of society.

4. Software Process Activities and Models

4.1 Selected Software Process Model

4.1 Selected Software Process Model

The team selected the Agile software process model for Citizen Care System. Agile is an iterative and incremental development approach that emphasizes flexibility, collaboration, continuous feedback, and rapid delivery of working software components.

activity	Application in Citizen Care System
requirement	Gathered from government service guidelines, user feedback on existing platforms, and health service needs
design	Creating UML diagrams and designing the system architecture
implementation	Each sprint (two weeks) delivers one working feature
testing	Security, performance, and usability testing happen in every sprint
Maintenance	Fixing bugs, releasing small updates, and keeping servers running

4.2 Justification for Model Selection

The team selected Agile and rejected the Waterfall model for the following reasons:

1. Changing Requirements: Citizen Care System must continuously evolve as government regulations and citizen needs change. New features such as vaccination campaigns, new appointment types, and emergency response updates can be incorporated in new sprints without disrupting the entire project.
2. Emergency Responsiveness: The system must respond to new emergency types and government directives quickly. Agile allows the team to react and deploy changes within a two-week sprint rather than waiting months for a new version.
3. User Feedback: The platform serves millions of users. Collecting feedback after each sprint and applying improvements every two weeks ensures the system stays aligned with actual user needs rather than assumptions made at the beginning of the project.
4. Continuous Security Testing: Since the system handles sensitive health data and personal identity information, security cannot be tested only at the end of the project. In Agile, security testing is integrated into every sprint, ensuring continuous protection throughout development.

The Waterfall model was rejected because:

- Requirements are not fully known at the beginning of the project
- New health and government needs emerge regularly throughout development
- Users cannot wait months for important features to be delivered
- Security testing must happen continuously and cannot be deferred to the final stage

5. Software Requirements (SRS)

5.1 System Requirements

The requirements for the Citizen Care System application were gathered by studying the main functions and services provided by the system. The development team analyzed how users interact with the application and identified the most important services that should be available to users. The requirements were collected based on user needs, system objectives, and the main functionalities presented in the Use Case Diagram.

Several methods were used during the requirements gathering process. First, the team reviewed similar electronic government and health service applications in order to understand the essential features that users expect from such systems. This helped the team identify the most common and important services that should be included in the application. Second, discussions and meetings were conducted among team members to analyze how the system should operate and how users would interact with the application in different situations.

In addition, the team focused on designing a system that is secure, simple, and reliable because the application manages sensitive personal and health-related information. The system was planned to ensure that users can access services safely while protecting their private data from unauthorized access. Security requirements such as secure login, identity verification, and data protection were considered important parts of the system design.

The collected requirements were organized into two main categories: functional requirements and non-functional requirements. Functional requirements describe the specific services and operations that the system provides to users, such as account registration, login, permit booking, viewing health status, accessing government services, updating personal information, receiving notifications, and viewing user history. These requirements explain what the system should do and how users interact with the application.

On the other hand, non-functional requirements describe the quality and performance of the system. These requirements focus on aspects such as security, usability, reliability, performance, and system availability. The application should be easy to use for all users, respond quickly to user requests, and remain available without interruptions. The system should also maintain accurate and updated information at all times.

The Citizen Care System application is designed to help users access government and health services electronically through a secure mobile application. The system aims to reduce the need for physical visits to government offices by allowing users to complete services digitally and efficiently. This helps save time and effort for users while improving the overall user experience.

Furthermore, the application is designed to provide users with quick access to important information and services through one unified platform. The notification service helps users stay informed about updates, reminders, permits, and announcements related to the services provided by the system. The application also supports easy navigation and organized service management to improve accessibility and convenience for users.

Overall, the requirements gathering process helped define the essential features and quality standards needed for the Citizen Care System. These requirements ensure that the system can provide secure, efficient, and user-friendly electronic services while meeting the needs of users and supporting digital government services effectively.

5.2 Requirements Engineering Plan

1. Requirements Gathering Methods

We gathered the system requirements using the following methods:

- **Reviewing existing applications:** We studied similar e-government and health service platforms such as Tawakkalna, Absher, and Sehaty to understand essential features expected by users.
- **Team brainstorming sessions:** We conducted three group meetings (1 hour each) where all members discussed and proposed possible services and features for the system.
- **User scenario analysis:** We identified five common user types (citizen, resident, visitor, health worker, system admin) and defined what each user would need from the system.

2. Requirements Analysis Methods

We analyzed the gathered requirements using a simple and clear approach:

- **Categorization:** Each requirement was labeled as either Functional (what the system does) or Non Functional (how well the system performs).
- **Prioritization:** We classified the functional requirements into three priority levels:

Requirements	Priority
Registration, Login, Display health status, Permit booking	Essential
Notifications, History tracking, Update personal profile	Important
Bilingual support (Arabic/English)	Future

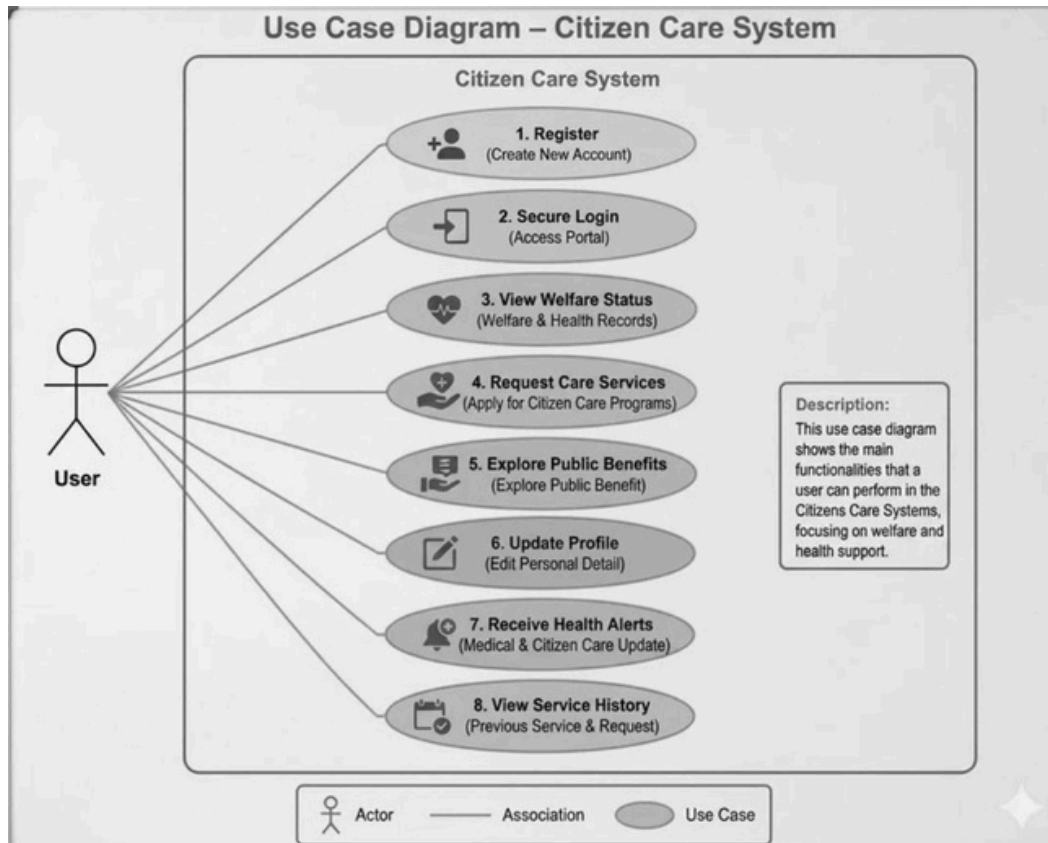
Mapping to use cases: We ensured that every functional requirement was represented in the Use Case Diagram to check completeness.

3. Requirements Validation Methods

We validated the requirements using these methods:

- **Team review:** All six team members reviewed the requirements list and confirmed that nothing important was missing.
- **Coverage check:** We verified that we have at least 8 functional requirements, 4 non-functional requirements, and 3 external interface requirements as required by the project assessment criteria.
- **Consistency check:** We ensured that no two requirements contradicted each other (for example, no requirement asking for both high security and no login).

5.3 Ues case diagram



5.4 Functional Requirements

The Citizen Care System provides several important functions that help users access government and health services easily, efficiently, and securely. These functions are considered the core services of the application and allow users to perform different tasks through one integrated digital platform. The system is designed to improve the user experience by reducing the need for physical visits to government offices and providing electronic access to important services anytime and anywhere.

One of the main functions of the system is user registration and authentication. The system allows users to create a new account by entering personal information and verifying their identity through secure authentication methods. This process ensures that only authorized users can access the application and its services. After successful registration, users can securely log into the system using their credentials. The login process is designed to maintain privacy and protect sensitive user information from unauthorized access.

Another important function of the application is displaying the user's health status. The system allows users to view health-related information such as vaccination records, immune status, and other medical information available within the application. This feature enables users to quickly access their health information whenever needed and supports digital health management through a centralized platform.

The application also provides an electronic permit booking service. Users can request and book permits directly through the system without needing to visit government offices physically. This service simplifies administrative procedures and helps users save both time and effort. The permit feature improves service efficiency and provides users with a faster and more convenient experience.

In addition, the Citizen Care System offers access to multiple government services through one unified application. Users can navigate through the system and access available services in an organized and user-friendly way. This integration of services allows users to complete different government-related tasks electronically using a single platform, which improves accessibility and service management.

The system also allows users to update and edit their personal profile information whenever necessary. Users can modify personal details such as contact information and account data to ensure that all stored information remains accurate and updated. Maintaining updated user information helps improve communication between the system and users and ensures better service delivery.

Furthermore, the application includes a notification and alert service that informs users about important updates, reminders, announcements, and service-related information. Notifications help users stay informed about the latest activities within the system, including permit approvals, health updates, and government announcements. This feature improves communication and ensures that users receive timely information.

Another key function of the system is the history and activity tracking feature. The application allows users to review their previous activities, permits, and services used within the system. Users can access their history records whenever needed, which helps them track completed services and maintain organized records of their activities.

The Citizen Care System is also designed with a strong focus on usability, reliability, and security. The application interface is simple and easy to navigate, allowing users from different backgrounds to access services without difficulty. In addition, the system aims to provide fast response times and continuous service availability to ensure a smooth and reliable user experience.

Security is considered one of the most important aspects of the system because the application manages sensitive personal and health-related information. Therefore, the system uses secure authentication and data protection mechanisms to maintain user privacy and ensure the confidentiality of stored information. The system also ensures that users can safely access services while protecting their personal data.

Overall, the Citizen Care System serves as an integrated electronic platform that combines government and health services in one secure application. The system improves accessibility, enhances service efficiency, and supports digital transformation by providing users with convenient and reliable electronic services through a modern mobile application.

5.5 Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance standards of the Citizen Care System. These requirements focus on how the system operates rather than the specific services it provides. They are important for ensuring system efficiency, reliability, security, and user satisfaction.

One of the most important non-functional requirements is security. Since the Citizen Care System handles sensitive personal and health information, the system must provide strong security measures to protect user data. Secure login methods, encrypted communication, and data protection mechanisms should be implemented to prevent unauthorized access and maintain user privacy.

The system should also provide high performance and fast response time. Users should be able to access services, move between pages, and receive information quickly without delays. Fast system performance improves the user experience and increases system efficiency during daily use.

Availability is another important requirement for the Citizen Care System. The system should be available 24 hours a day and 7 days a week with minimal downtime. Continuous availability is necessary because users may need to access health information, permits, or government services at any time.

The application should also be easy to use and understandable for all users. The interface should be simple, organized, and user-friendly so that people with different technical skills can use the application easily. Supporting both Arabic and English languages also improves accessibility and usability for different users.

In addition, the system should provide reliability and stability while handling large numbers of users at the same time. The application must continue operating correctly even during busy periods or high system usage without crashing or slowing down significantly.

The Citizen Care System should also maintain compatibility with different mobile devices and operating systems. The application must function properly on smartphones and tablets to ensure that users can access services from different devices without technical issues.

Overall, the non-functional requirements help ensure that the Citizen Care System provides secure, stable, fast, and user-friendly services while maintaining high system quality and performance.

5.6 External Interface Requirements.

External interface requirements describe the communication and interaction between the Citizen Care System, users, external systems, and communication networks. These requirements are important because they help the application operate smoothly and allow users to access services easily and securely. The external interfaces also improve the overall user experience and support communication between the system and other government platforms

The Citizen Care System application provides a user-friendly interface that allows users to interact with the system in a simple and organized way. The interface includes several important screens ,and services such as account registration, secure login, health status display, permit booking notifications, and profile management. The design of the application should be clear and easy to understand so that users of different ages and technical backgrounds can use the system without difficulty. In addition, the interface should support both Arabic and English languages to improve .accessibility for all users in Saudi Arabia

The application should also provide a responsive interface that works properly on different mobile devices and screen sizes. Since the Citizen Care System is mainly used as a mobile application, the .system should maintain consistent performance and display quality on smartphones and tablets The interface elements such as buttons, menus, and notifications should be organized clearly to .help users navigate through the application efficiently

Furthermore, the Citizen Care System communicates with several external governmental systems to provide integrated services. The application connects with the Absher platform and the .National Single Sign-On (SSO) system to verify user identities and allow secure authentication The system also communicates with Ministry of Health services to display health status information, vaccination records, and other health-related data. These software interfaces help .ensure that the information displayed in the application is accurate, updated, and reliable

The integration between the Citizen Care System and external government systems is essential because it allows users to access multiple services through one application without needing to use separate systems. This integration improves efficiency and simplifies the process of accessing .government and health services electronically

In addition, the application depends on internet connectivity to send and receive information between users and the system. Users can access the application using Wi-Fi networks or mobile internet services such as 4G and 5G. The communication interface must provide stable and secure .connections to ensure that services operate correctly without interruptions

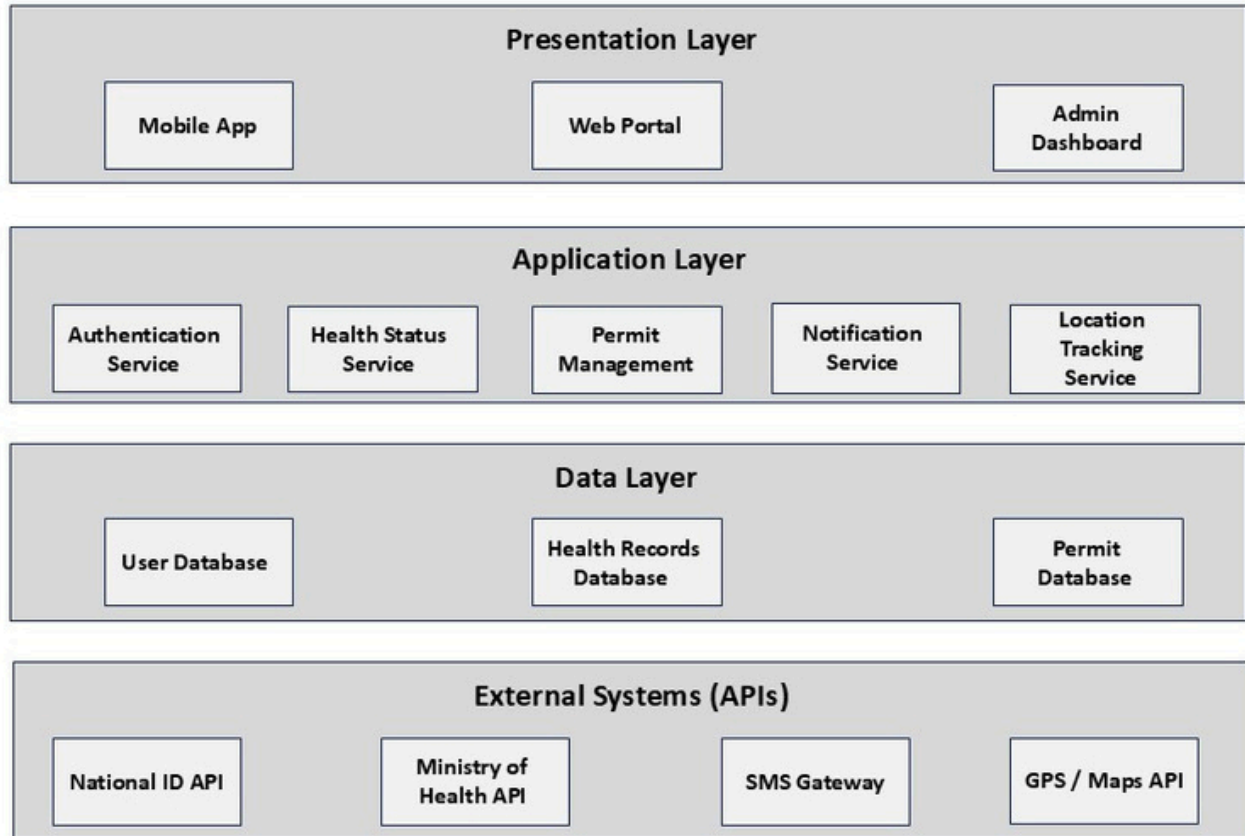
The system should also use secure communication protocols such as HTTPS to protect data transferred between the user device and the system servers. This is important because the application handles sensitive information such as personal data, health records, and official .documents. Secure communication helps prevent unauthorized access and protects user privacy

Moreover, the application should support push notification services to deliver alerts and updates ,directly to users. Notifications may include reminders, important announcements, permit updates and service-related information. The notification system helps users stay informed and improves .communication between the system and users

Overall, the external interface requirements help ensure that the Citizen Care System provides secure communication, reliable system integration, and an easy user experience. These interfaces play an important role in maintaining system efficiency, supporting government services, and .improving user satisfaction

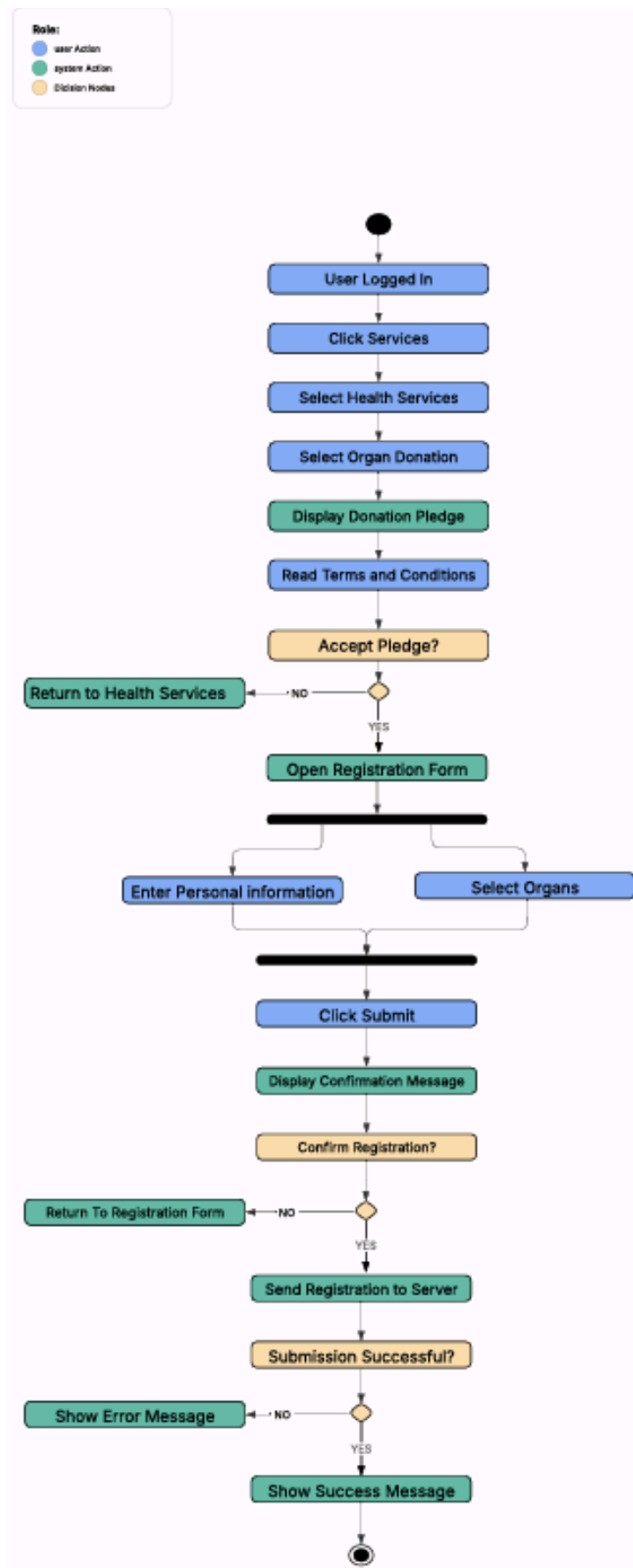
6. System/Software Modeling (SDD)

6.1 Architecture Model



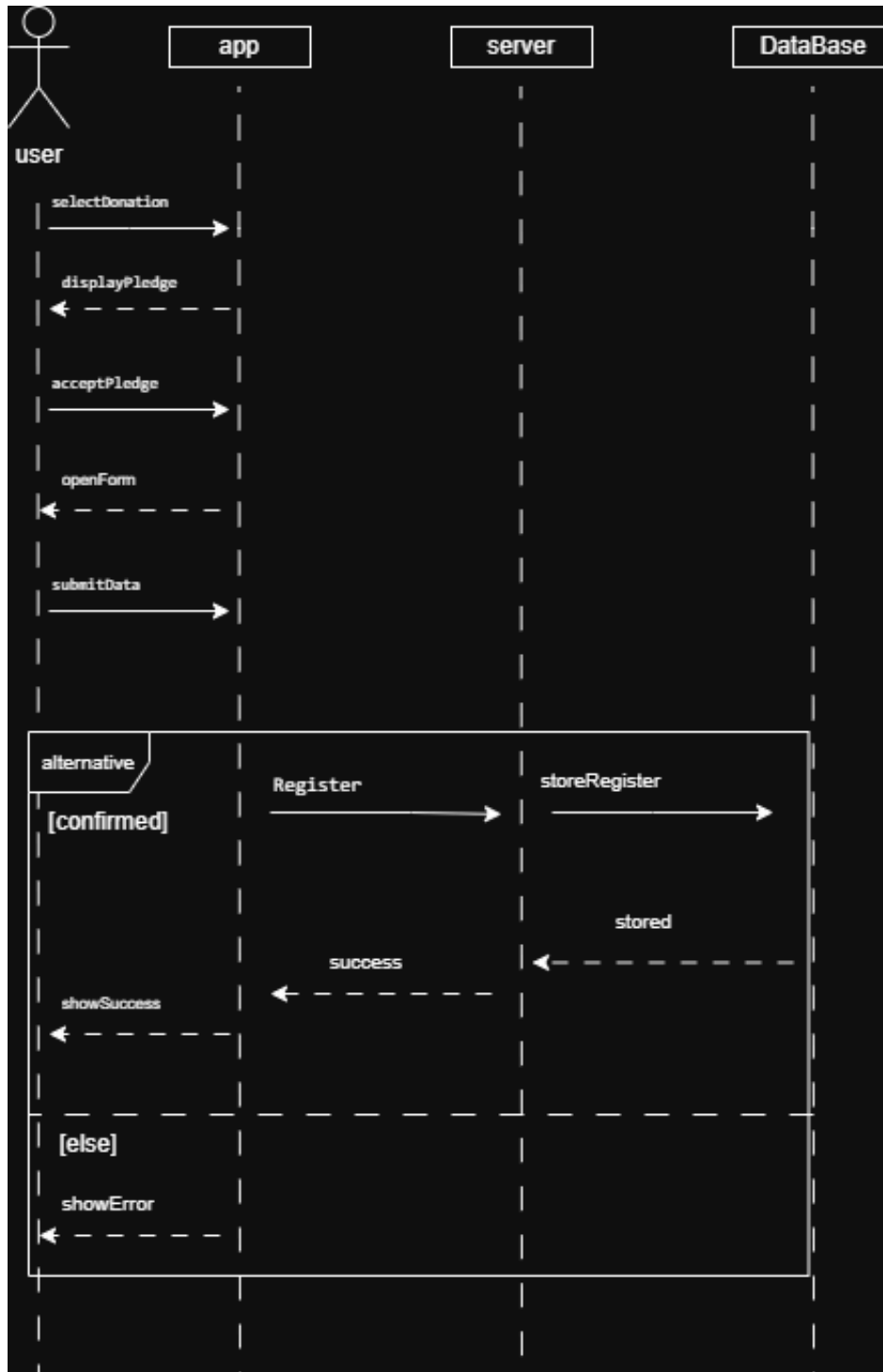
6.2 UML Diagrams Activity diagram

[Click Here to Access the diagram with better resolution](#)

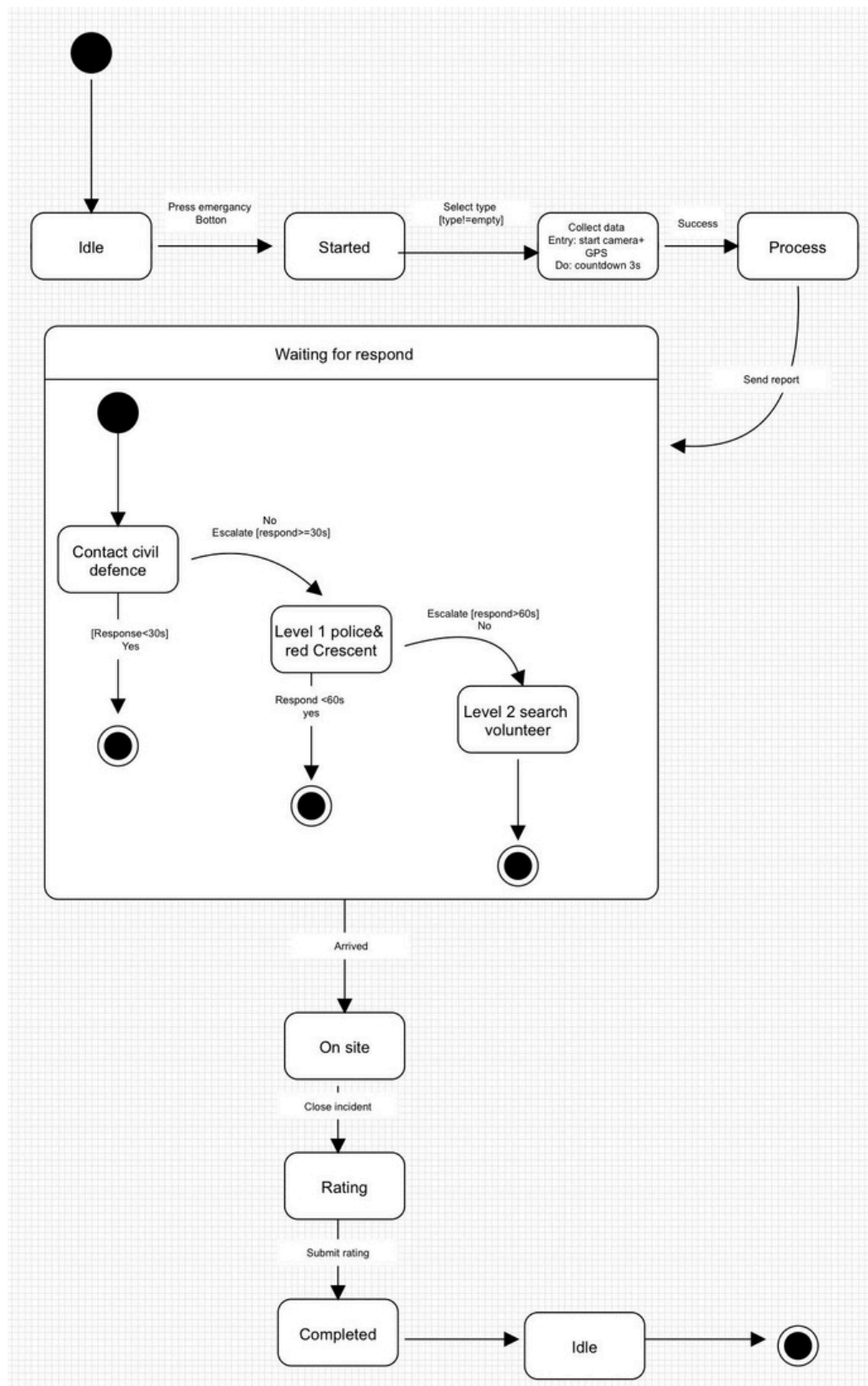


6.3 UML Diagrams Sequence Diagram

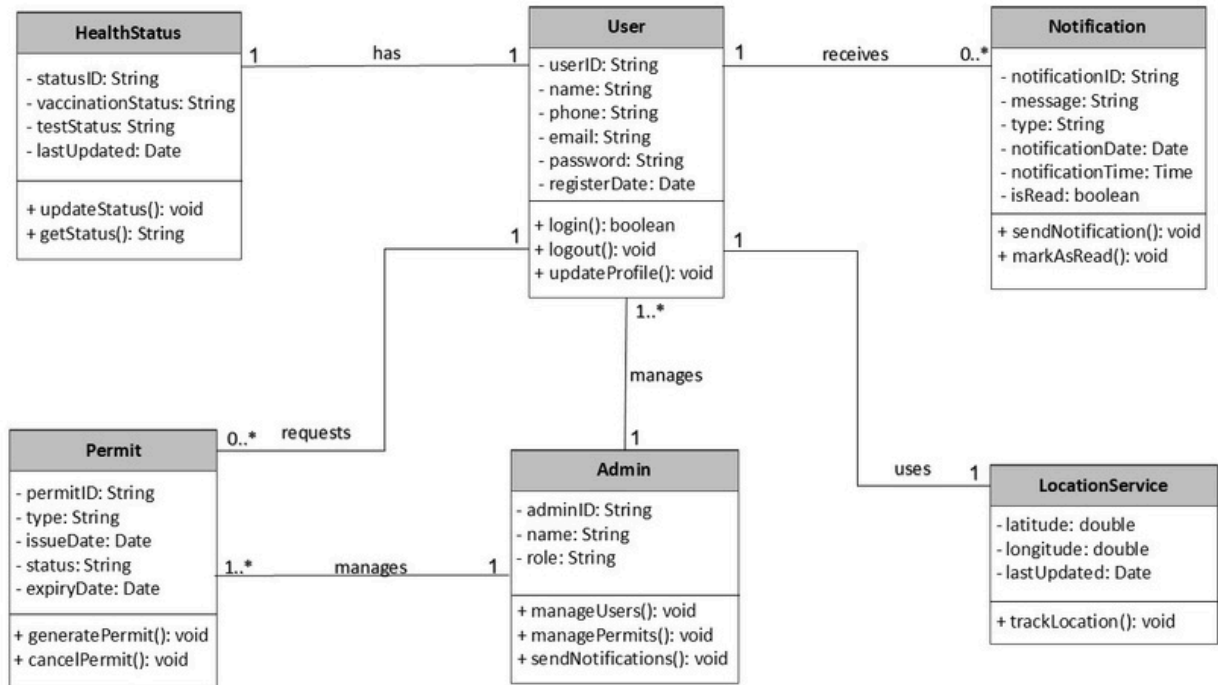
[diagram link](#)



6.4 UML Diagrams State Diagram



6.5 UML Diagrams class Diagram



7.Contributions

Name	ID	Contribution
Nada Alharbi	432205291	Software Engineering Ethics and Responsibility , meeting agenda(4)
Leen Almansour	432205435	UML Diagrams State Diagram,Requirements Engineering Plan writing,Software Process Activities and Models , meeting agenda(1)
Norah Almrzooq	441203264	Architecture Model Design, UML Class Diagram, Slide Formatting.
Dhiaa Alharbi	432205322	UML Activity diagram, UML sequence diagram , introduction
Gader Alquniha	441203382	Non-Functional Requirements,External Interface Requirements ,Meeting Agenda(2)
Rahma Alsuhaymi	432205312	System Requirements Functional Requirements Use case diagram Meeting Agenda (3)

8.1 Meeting Agenda (1)

DAY :SATRDAY
DATE: 14.3.2026
TIME: 12:00 AM - 01:00 AM
LOCATION : ZOOM

Agenda Items:

1. Welcome and introductions
2. Discuss potential project topics
3. Select final project topic
4. Decide when to start working
5. Divide tasks among team members
6. Agree on meeting schedule and communication tools

Expected Outcomes:

- Project topic confirmed
- Work start date set
- Initial task distribution

Attendees:Nada Alharbi-Leen Almansour-Norah Almrzooq-Dhiala Alharbi-
Gader Alquniha-Rahma Alsuhaymi
agenda prepared by: leen almansour

Meeting Minutes – Session 1

Decisions made:

- Final project topic: Citizen Care System+ (government services platform)
- Work will start immediately after the meeting
- Communication tool: WhatsApp group + Canva for documentation
- Meeting schedule: Every 3-4 weeks (next meeting: 15.4.2026)

Tasks assigned with deadlines:

Member Task Deadline

Nada Alharbi Team leader, coordinate work Ongoing
Leen Almansour Prepare agenda template, UML State Diagram 20.4.2026
Norah Almrzooq Architecture model, Class Diagram 20.4.2026
Dhiala Alharbi Activity Diagram, Sequence Diagram, Introduction 20.4.2026
Gader Alquniha Non-functional requirements, External interface 20.4.2026
Rahma Alsuhaymi System requirements, Functional requirements, Use Case 20.4.2026

Follow-up from previous meeting: N/A (first meeting)

8.2 Meeting Agenda (2)

DAY : FRIDAY
DATE: 15.4.2026
TIME: 02:00 PM - 03:00 PM
LOCATION : ZOOM

Agenda Items:

1. First review of completed work
2. Check for errors in each section
3. Discuss corrections needed
4. Review UML diagrams for accuracy
5. Confirm Software Process Model section

Expected Outcomes:

- List of corrections identified
- Updated tasks assigned

Attendees: Nada Alharbi-Leen Almansour-Norah Almrzooq-Dhiaa Alharbi-Gader Alquniha-Rahma Alsuhaymi

Agenda prepared by: Gader Alquniha

Meeting Minutes – Session 2

Decisions made:

- Agile model is confirmed and justified
- All UML diagrams
- External interface requirements need more details
- Use Case diagram needs to be linked to functional requirements

Tasks assigned with deadlines:

Member Task Deadline

Gader Alquniha Expand external interface requirements section 25.4.2026

Rahma Alsuhaymi Map use cases to functional requirements 25.4.2026

Dhiaa Alharbi Fix sequence diagram symbols 22.4.2026

Follow-up from previous meeting (Session 1):

- All tasks from Session 1 were completed on time
- No pending tasks

8.3 Meeting Agenda (3)

DAY : Sunday
DATE:17. 5.2026
TIME: 07:00 PM - 08:00 PM
LOCATION : ZOOM

Agenda Items:

1. Discuss work progress
2. Final check of all content
3. Ensure all criteria are met (8 functional, 4 non-functional, etc.)
4. Prepare final document for submission
5. Upload content to repository

Expected Outcomes:

- Final document ready
- Repository organized

Attendees:Nada Alharbi-Leen Almansour-Norah Almrzooq-Dhiaa Alharbi-Gader Alquniha-Rahma Alsuhaymi

Agenda prepared by: Rahma Alsuhaymi

Meeting Minutes – Session 3.

Decisions made:

- All content is complete and ready for submission
- Repository structure is organized
- Meeting agenda and minutes will be added to the report

Tasks assigned with deadlines:

Member Task Deadline

All members Final review of their sections 20.5.2026

Leen Almansour Compile final PDF report 25.5.2026

Norah Almrzooq Check slide formatting 25.5.2026

Follow-up from previous meeting (Session 2):

All correction tasks from Session 2 were completed

8.4 Meeting Agenda (4)

DAY : Tuesday
DATE: 19.5.2026
TIME: 06:00 PM - 07:00 PM
LOCATION : ZOOM

Agenda Items:

- 1,final explanations and training
- 2, practice. Presentation
- 3, each member explains thier assigned part
- 4, Time managementrehearsal
- 5, preparefor Q&A session

Expected Outcomes:

- All members are fully prepared for the final presentation.
- Total presentation time is successfully controlled within the allowed limit.

Attendees:Nada Alharbi-Leen Almansour-Norah Almrzooq-Dhiaa Alharbi-Gader Alquniha-Rahma
Alsuhaymi
Agenda prepared by:Nada Alharbi

Meeting Minutes – Session 4

Decisions made:

- The final presentation order among members is confirmed.
- Each member will speak for approximately 1.5 to 2 minutes.
- Total presentation time is optimized to be within 10 minutes.
- Presentation slides are officially finalized, polished, and ready.

Tasks assigned with deadlines:

- **Task 1:** All members must practice their assigned parts individually at home. **Deadline:** Presentation Day.
- **Task 2:** The project team must prepare expected Q&A answers and technical justifications. **Deadline:** Presentation Day.

Follow-up from previous meeting (Session 3):

- Final report is completed and submitted.
- Project repository is accessible.

References

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yes case diagram used artificial intelligence to assist in creating the drawing
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External interface requirements–“YouTube”